A.

$\chi^2_{\text{Kruskal–Wallis}}(3) = 71.00$, $p = 2.61\text{e}{-15}$, $\hat{\varepsilon}^2_{\text{ordinal}} = 0.06$, $\text{CI}_{95\%}$ [0.04, 1.00], $n_{\text{obs}} = 1{,}232$

$p_{\text{Holm–adj.}} = 6.37\text{e}{-08}$
$p_{\text{Holm–adj.}} = 1.32\text{e}{-12}$
$p_{\text{Holm–adj.}} = 4.59\text{e}{-14}$
$p_{\text{Holm–adj.}} = 0.03$
$p_{\text{Holm–adj.}} = 2.80\text{e}{-03}$

Alpha diversity values (Shannon)

0
2
4
6

Milk Grain
Milk Liquid
Water Grain
Water Liquid

Pairwise test: **Dunn**, Bars shown: **significant**

Household conditions

B.

$\chi^2_{\text{Kruskal–Wallis}}(2) = 58.48$, $p = 2.00\text{e}{-13}$, $\hat{\varepsilon}^2_{\text{ordinal}} = 0.06$, $\text{CI}_{95\%}$ [0.04, 1.00], $n_{\text{obs}} = 1{,}016$

$p_{\text{Holm–adj.}} = 1.31\text{e}{-13}$
$p_{\text{Holm–adj.}} = 2.85\text{e}{-04}$
$p_{\text{Holm–adj.}} = 0.03$

Alpha diversity values (Shannon)

0
2
4
6

Milk Liquid
Milk Liquid
Water Liquid

Pairwise test: **Dunn**, Bars shown: **significant**

Laboratory controlled
Household conditions

C

Kefir type

Fermentation conditions

Milk Grain

Paenibacillus odorifer
Leuconostoc mesenteroides
Lentilactobacillus hilgardii
Lentilactobacillus buchneri
Lactococcus garvieae
Lactobacillus acidophilus
Lacticaseibacillus paracasei
Klebsiella quasipneumoniae
Klebsiella pneumoniae
Klebsiella michiganensis
Klebsiella grimontii
Escherichia coli
Enterobacter hormaechei
Clostridium botulinum
Clostridium beijerinckii
Brettanomyces bruxellensis

wk1 wk5 wk9 wk13 wk17 wk21

Water Grain

Schleiferilactobacillus harbinensis
Rhizobium pusense
Pantoea vagans
Oenococcus oeni
Limosilactobacillus fermentum
Leuconostoc gelidum
Klebsiella pneumoniae
Delftia lacustris
Delftia acidovorans
Comamonas testosteroni
Carnobacterium maltaromaticum
Candidatus Sulcia muelleri
Brochothrix thermosphacta
Agrobacterium tumefaciens
Acinetobacter ursingii

wk1 wk5 wk9 wk13 wk17 wk21

Species

ML

Paenibacillus odorifer
Leuconostoc mesenteroides
Lentilactobacillus buchneri
Lactococcus garvieae
Lacticaseibacillus paracasei
Klebsiella quasipneumoniae
Klebsiella pneumoniae
Klebsiella grimontii
Escherichia coli
Enterobacter hormaechei
Clostridium beijerinckii
Carnobacterium maltaromaticum
Brettanomyces bruxellensis

wk1 wk5 wk9 wk13 wk17 wk21

WL

Schleiferilactobacillus harbinensis
Rhizobium pusense
Pantoea vagans
Oenococcus oeni
Limosilactobacillus fermentum
Leuconostoc gelidum
Klebsiella pneumoniae
Klebsiella grimontii
Comamonas testosteroni
Carnobacterium maltaromaticum
Brochothrix thermosphacta
Agrobacterium tumefaciens
Acinetobacter ursingii

wk1 wk5 wk9 wk13 wk17 wk21

Relative abundance (%)

10 20 30 40

Number of participants

1 4 7 10 13

10 20 30 40